

Introduction

The requirements defined for this project were defined following a MoSCoW approach, which categorizes the requirements into:

- Must have (M) - critical requirements that are non-negotiable and need to be present in the product;
- Should have (S) - important, high-value requirements that should be included if possible, even if they are not present in a first "release";
- Could have (C) - desirable, "nice to have" requirements that improve the product but won't really affect its value if not present;
- Won't have (W) - requirements agreed that are not a priority for the current time frame.

Requirements resumed table

ID	Requirement	Pipeline	Priority
FR0.1	Load Historical data	Base system	M
M1- FR1.1	Upload of clinical trial	Conversion and Application of Clinical Trial Criteria	S
M1- FR1.2	Selection of clinical trial	Conversion and Application of Clinical Trial Criteria	S
M1- FR1.3	Pre-processing of clinical trial	Conversion and Application of Clinical Trial Criteria	S
M1- FR1.4	Extract clinical trial criteria	Conversion and Application of Clinical Trial Criteria	M
M1- FR1.5	Convert criteria into logic	Conversion and Application of Clinical Trial Criteria	M
M1- FR1.6	Presentation of rules	Conversion and Application of Clinical Trial Criteria	M
M1- FR1.7	Validation of inclusion and exclusion criteria	Conversion and Application of Clinical Trial Criteria	M
M1- FR1.8	Persistence of rules	Conversion and Application of Clinical Trial Criteria	M
M2- FR1.1	Matching of patients	Conversion and Application of Clinical Trial Criteria	M
M2- FR1.2	Generate eligibility report	Conversion and Application of Clinical Trial Criteria	S

ID	Requirement	Pipeline	Priority
M2-FR1.3	Justification generation	Conversion and Application of Clinical Trial Criteria	S
M2-FR1.4	Persistence of matching	Conversion and Application of Clinical Trial Criteria	M
DR1.1	Criteria mapping	Conversion and Application of Clinical Trial Criteria	M
IR1.1	Clinical trial submission	Conversion and Application of Clinical Trial Criteria	C
IR1.2	Clinical trial selection	Conversion and Application of Clinical Trial Criteria	S
IR1.3	Reporting interface	Conversion and Application of Clinical Trial Criteria	C
IR1.4	Clinical Trial Validation interface	Conversion and Application of Clinical Trial Criteria	S
FR2.1	Selection of clinical trial	Monitoring the Quality of Care	S
FR2.2	Retrieve treatment data	Monitoring the Quality of Care	M
FR2.3	Compute OS and PFS	Monitoring the Quality of Care	M
FR2.4	Apply Kaplan-Meier analysis	Monitoring the Quality of Care	M
FR2.5	Visualize results	Monitoring the Quality of Care	S
DR2.1	Required data fields	Monitoring the Quality of Care	M
DR2.2	Derived variables	Monitoring the Quality of Care	M
IR2.1	Analysis dashboard	Monitoring the Quality of Care	S
IR2.2	Report export	Monitoring the Quality of Care	C
FR3.1	Select Diaries for extraction	Hyper-Relevant Parameter Extraction	C
FR3.2	Pre-processing of clinical diary	Hyper-Relevant Parameter Extraction	S
FR3.3	Extract structured info	Hyper-Relevant Parameter Extraction	M
FR3.4	Normalize text	Hyper-Relevant Parameter Extraction	M
FR3.5	Presentation of info	Hyper-Relevant Parameter Extraction	M
FR3.6	Validation of extracted parameters	Hyper-Relevant Parameter Extraction	M

ID	Requirement	Pipeline	Priority
FR3.7	Store structured profiles	Hyper-Relevant Parameter Extraction	M
DR3.1	Extracted fields	Hyper-Relevant Parameter Extraction	M
DR3.2	Storage schema	Hyper-Relevant Parameter Extraction	M
IR3.1	Clinical diary page	Hyper-Relevant Parameter Extraction	C
IR3.2	Clinical diary selection	Hyper-Relevant Parameter Extraction	S
IR3.3	Clinical Diary Validation interface	Hyper-Relevant Parameter Extraction	S

Sea Level Requirements

Requirements for the system

ID	Requirement	Description	Priority
FR0.1	Load historical data	The clinical trials since (a year) should be loaded into the system in order for it to start working	M

Identification of Eligible Patients for Clinical Trials

Different from the following components this one has two modules:

- **Extraction and Conversion of Criteria** - as the name suggests this first module will be responsible for the extraction of inclusion/exclusion criteria and its conversion into logical rules.
- **Deterministic Matching of Patients** - this last module is responsible for the matching of patients to the selected clinical trial, this process will be based on the rules extracted on the previous step and will retrieve only the already structured patients' profiles persisted on the database

Functional Requirements

Extraction and Conversion of Criteria

ID	Requirement	Description	Priority
M1-FR1.1	Upload of clinical trial	The system must allow the upload of clinical trials that can be manipulated on following steps	S
M1-FR1.2	Selection of clinical trial	The system must allow the selection of a clinical trial to be submitted to the pipeline of conversion and matching of patients	S
M1-FR1.3	Pre-processing of clinical trial	The system must perform a pre-processing step for the removal of unnecessary information (headers/footers, sections that are not the inclusion/exclusion criteria)	S
M1-FR1.4	Extract clinical trial criteria	The LLM must extract inclusion and exclusion criteria from a clinical trial	M
M1-FR1.5	Convert criteria into logic	The extracted text must be converted into logical expressions (e.g., Age > 18), this conversion must follow the schema enforced by the database	M
M1-FR1.6	Presentation of rules	After the extraction and conversion of the criteria into logical rules, these must be presented to the doctor for further validation	M
M1-FR1.7	Validation of inclusion and exclusion criteria	After the LLM extracts the criteria of inclusion and exclusion a second phase must be undergone, the doctor must validate this results in order for the patients to be elected.	M
M1-FR1.8	Persistence of rules	After confirming and validating the logical rules, these must be persisted on the database	M

Deterministic Matching of Patients

ID	Requirement	Description	Priority
M2-FR1.1	Matching of patients	The system must deterministically evaluate patient data against the logical rules that were persisted on the previous steps, a binary decision must be outputted	M
M2-FR1.2	Generate eligibility report	The system must output a list of eligible and excluded patients	S
M2-FR1.3	Justification generation	Appended to the eligibility report must be a justification for each patient decision, justifying the decision and criteria utilized for it, these justifications will be on Natural Language generated by an LLM	S

ID	Requirement	Description	Priority
M2-FR1.4	Persistence of matching	After a patient is considered eligible for a trial, he should be paired with it and this information must be persisted on the database	M

Data Requirements

ID	Requirement	Description	Priority
DR1.1	Criteria mapping	Each criterion must map to corresponding patient attributes in the database	M

Interface Requirements

ID	Requirement	Description	Priority
IR1.1	Clinical trial submission	The clinician must be able to submit several clinical trials into the system.	C
IR1.2	Clinical trial selection	The clinician must be able to select a clinical trial and view eligibility results.	S
IR1.3	Reporting interface	The system must display the justification report in a readable format (PDF/HTML).	C
IR1.4	Clinical Trial Validation interface	A page must be displayed, allowing for the doctor to edit and validate the criteria extracted	S

Monitoring the Quality of Care

Functional Requirements

ID	Requirement	Description	Priority
FR2.1	Selection of clinical trial	The system must allow the selection of a clinical trial for the survivability estimation calculation	S
FR2.2	Retrieve treatment data	The system must retrieve patient treatment records: - Diagnosis date; - First cycle date - Progression date/End of treatment; - Death date; - Status (0/1): - 0: censored (patient didn't had the event until	M

ID	Requirement	Description	Priority
		the end of the study); - 1: event occurred (progression or death).	
FR2.3	Compute OS and PFS	The system must calculate Overall Survival (OS) and Progression-Free Survival (PFS).	M
FR2.4	Apply Kaplan-Meier analysis	The system must use Kaplan-Meier survival estimation to analyze outcomes.	M
FR2.5	Visualize results	The system must generate survival curves and statistical summaries.	S

For the requirement FR2.1, the extracted fields from the "Automated Structured Data Extraction" phase are mapped with the necessary fields for calculating the Kaplan-Meier:

Extracted field	Necessary field	obs
Date of first treatment	First cycle date	-
Diagnosis date	Diagnosis date	-
Death date	Death date	If it exists for OS event = 1 else event = 0
End date (from treatments)	End of treatment	For PFS event = 0
-	Progression date	For PFS event = 1

Data Requirements

ID	Requirement	Description	Priority
DR2.1	Required data fields	Diagnosis date, first cycle date, progression/end date, death date, and status (0/1).	M
DR2.2	Derived variables	OS = (death date – diagnosis date); PFS = (progression date – first cycle date).	M

Interface Requirements

ID	Requirement	Description	Priority
IR2.1	Analysis dashboard	The clinician must be able to select patient cohorts and view survival graphs.	S
IR2.2	Report export	Results should be exportable to CSV, PDF, or compatible with SPSS.	C

Automated Structured Data Extraction

Functional Requirements

ID	Requirement	Description	Priority
FR3.1	Select Diaries for extraction	The system must allow one or multiple diaries to be extracted	C
FR3.2	Pre-processing of clinical diary	The system must perform a pre-processing step for the removal of unnecessary information to the pipeline (headers/footers, etc...)	S
FR3.3	Extract structured info	The LLM must extract patient data from free-text diaries: - Age / date of birth; - ECOG-PS; describes a patient's level of functioning in terms of their ability to care for themselves, daily activity, and physical ability; - Diagnosis; - Date of Diagnosis; - Molecular Status; - Stage; - Date of Treatment; - Treatments (First, Second); ? protocolo de quimio? - Start Date; - End Date; - Control In case of fields not existing on the clinical diary the field must be defined to None.	M
FR3.4	Normalize text	The system must correct typographical and linguistic inconsistencies.	M
FR3.5	Presentation of info	After the extraction and normalization steps this information must be presented to the doctor	M
FR3.6	Validation of extracted parameters	After the LLM extracts the parameters the doctor can edit and validate these data.	M
FR3.7	Store structured profiles	Each structured patient record must be stored in the local database.	M

Data Requirements

ID	Requirement	Description	Priority
DR3.1	Extracted fields	Age / date of birth, ECOG-PS, Diagnosis, Date of Diagnosis, Stage, Molecular Status, Date of Treatment, Treatments, Treatment start date and Treatment end date and control information	M
DR3.2	Storage schema	All extracted data must be stored on a postgresql database	M

Interface requirements

ID	Requirement	Description	Priority
IR3.1	Clinical diary page	The clinician must be able to list all the unstructured clinical diaries in the system	C
IR3.2	Clinical diary selection	The clinician must be able to select a clinical diary and view eligibility results.	S
IR3.3	Clinical Diary Validation interface	A page must be displayed, allowing for the doctor to edit and validate the parameter extracted	S

Oyster Level Requirements

FR0.1 - Load historical data

All available unstructured clinical diary entries on SClinico for the oncology must be loaded into the PostgreSQL Database.

This will be done through a full load python script where unstructured data is read and stored in the a dedicated table on the DB. Because of the sheer amount of data it is unfeasible for it to be a manual process.

MAKE THE ZERO TRUST DATA LAKE ENVIRONMENT THE DB FOR THIS UNSTRUCTURED DATA.

FROM WHEN IS THIS DATA STORED.

PERIODICITY OF REFRESH ON DATALAKE.

DEFINE POPULATION DEFINITION (E.G. ONCOLOGY PATIENTS FROM 3 YEARS AGO)

POSTGRESQL DATABASE MUST BE INSIDE THE ZERO TRUST ENVIRONMENT.

M1-FR1.1 - Upload of clinical trial

The system must allow the upload of clinical trials that can be manipulated on following steps, this process must:

- Provide an interface that allows the visualization of already submitted clinical trials
- Allow the upload of PDF and TXT files into the system
- After the upload of the clinical trial this raw version must be persisted on the database for further use, creating a new father reference that will be a foreign key to all changes made to this file on downstream processes.

M1-FR1.2 - Selection of clinical trial

The system must allow the selection of a clinical trial to be submitted to the pipeline of conversion and matching of patients. This process must:

- Provide an interface that presents all clinical trials already submitted
- Allow the selection of one and only one clinical trial to be submitted to the conversion pipeline

M1-FR1.3 - Pre-processing of clinical trial

After the selection of a clinical trial, the system must perform a pre-processing step for the removal of unnecessary information. This process must:

- Remove any unnecessary information to the pipeline, in the case of clinical trials this is: headers of the file, footers of the file, sections that are not the inclusion/exclusion sections.
- After the end of this preprocessing step a new entry must be persisted on the database referencing the parent entry, making use of the foreign key.

M1-FR1.4 - Extract clinical trial criteria

The LLM must extract inclusion and exclusion criteria from a clinical trial. This process must:

- Parse the natural language text of the clinical trial
- Identify all eligibility criteria
 - Inclusion Criteria - under the "Inclusion Criteria" section on the clinical trial
 - Exclusion Criteria - under the "Exclusion Criteria" section on the clinical trial
 - If by any chances these sections aren't present the LLM must be capable of classifying the eligibility criteria into the respective categories.
 - A template for the prompt to guide the LLM is:

You are an assistant responsible for extracting eligibility criteria from a clinical trial.

Your task is to read the clinical trial text and produce a structured list of all eligibility criteria, separated into Inclusion Criteria and Exclusion Criteria.

Extraction Requirements:

- Parse the natural language text of the clinical trial.
- Identify all eligibility criteria.
- Use the "Inclusion Criteria" and "Exclusion Criteria" sections when they are explicitly present.
- If these sections are not present, classify each criterion into Inclusion or Exclusion based on its meaning and intent.
- Preserve the original meaning of each criterion without rewriting or interpreting it.
- Separate criteria into individual items. If a criterion contains multiple independent conditions, split them into separate entries.
- Do not remove details or add missing information.
- Output only the extracted criteria in the required structure.

Output Format:

```
{
  "inclusion_criteria": [
    "<criteria 1>",
    "<criteria 2>",
    ...
  ],
  "exclusion_criteria": [
    "<criteria 1>",
    "<criteria 2>",
    ...
  ]
}
```

Clinical trial to process:

```
<TRIAL>
{{TRIAL_TEXT}}
</TRIAL>
```

- With this natural language criteria a new entry must be persisted referencing the parent entry, making use of the foreign key.

M1-FR1.5 - Convert criteria into logic

After extracting inclusion and exclusion criteria from the clinical trial text (FR1.1), the system must use the LLM to convert each criterion into a structured, machine-readable logical representation. This process must:

- Transform free-text criteria into logical rules (e.g. comparison operators, Boolean conditions)
- This step must output the result in a standardized JSON logic format and the database schema must be enforced to avoid any incompatibilities, and so the following prompt was defined:

You are a clinical trial criteria logic-conversion assistant.

Your task is to transform free-text clinical trial inclusion and exclusion criteria into a structured, machine-readable logical representation that is strictly compatible with the patient database schema.

General Rules

- Convert only information explicitly stated in the criterion.
- Do not infer, assume, normalize, or guess missing information.
- If a criterion cannot be expressed deterministically using the allowed fields and operators, set `"logic": null`.
- Output only valid JSON matching the schema below.
- Do not include explanations, comments, or natural language.

Allowed Database Fields (FIELD)

You may only use the following fields:

- ``age``
- ``sex``
- ``diagnosis``
- ``stage``
- ``ecog_ps``
- ``molecular_status``
- ``treatment``
- ``diagnosis_date``
- ``treatment_start_date``
- ``treatment_end_date``
- ``progression_date``
- ``death_date``
- ``control``

If the criterion references a concept not present in this list, set `"logic": null`.

Allowed Operators (OPERATOR)

- ``=``

```
- `!=`  
- `>`  
- `>=`  
- `<`  
- `<=`  
- `IN`  
- `NOT IN`  
- `AND`  
- `OR`
```

No other operators are permitted.

Values (VALUE)

- Use only literal values explicitly stated in the criterion.
- Do not add units, ranges, or clinical interpretations.
- Dates must be represented as raw values if explicitly present.
- If a value is implicit or vague, the logic must be set to ``null``.

Logic Construction Rules

- If a criterion contains multiple conditions, combine them using explicit Boolean operators (``AND``, ``OR``).
- Boolean precedence must be explicit using parentheses.
- The logic expression must be fully evaluable against structured patient data.
- If any part of the criterion is non-computable, ambiguous, or subjective, set ``"logic": null``.

Output Schema

```
```json  
{
 "criterion": {
 "type": "inclusion" | "exclusion",
 "logic": string | null
 }
}
```

- This conversion must be stored in the database with a new entry, referencing the parent entry, making use of the foreign key.

## M1-FR1.6 - Presentation of rules

After the extraction of criteria (M1-FR1.4) and their conversion into logical rules (FR1.5), the system must present these results to the clinician for review. This process must:

- Display every inclusion and exclusion criterion extracted from the clinical trial text.
- For each criterion, show:
  - The original natural-language text.
  - The corresponding logical rule generated by the LLM.
- Ensure a clear one-to-one mapping between text and logic to support clinician understanding.

This process allows for a thorough validation of the rules.

## **M1-FR1.7 - Validation of inclusion and exclusion criteria**

Following the presentation of criteria and their logical rules (M1-FR1.6), the system must provide a dedicated interface for clinicians to validate and correct the extracted information. This process must:

- Allow the clinician to:
  - Review each inclusion and exclusion criterion previously presented.
  - Edit incorrect or ambiguous natural-language text extracted from the clinical trial.
  - Modify or correct the associated logical rule (fields, operators, values, Boolean structure).
- Require that all criteria for a given trial be explicitly validated before the patient-trial matching process (M2- FR1.1) can be executed.

## **M1-FR.8 - Persistence of rules**

After confirming and validating the logical rules, these must be persisted on the database. The process must:

- Persist these rules alongside the previous alterations to the clinical trial data.
- These rules must reference the parent entry, making use of the foreign key.

This process will allow clear traceability and makes it completely auditable.

## **M2-FR1.1 - Matching of patients**

For a given clinical trial, the system must recover and evaluate patient structured data against the logical criteria of clinical trials produced in FR1.2. This process must:

- Retrieve all structured patient records from the PostgreSQL database;
- Retrieve all logical eligibility criteria associated with the selected clinical trial from the `trial_criteria_logic` table.;
- Deterministically evaluate each patient against every inclusion and exclusion rule:
  - Inclusion criteria must all be satisfied;

- Exclusion criteria must not be satisfied;
- The evaluation must not rely on randomness or LLM inference at this stage.
- Generate an eligibility decision for each patient:
  - 1 - patient satisfies all inclusion criteria and violates none of the exclusion criteria;
  - 0 - otherwise.

## M2-FR1.2 - Generate eligibility report

After the result of M1-FR1.1 the system must output a list of eligible/excluded patients and LLM justification for each decision. The process must be done by:

- Request a justification for the decision from an LLM, having in account:
  - Inclusion and exclusion criteria;
  - Structured fields of the patient;
  - Deterministic decision result.
- Produce a structured dictionary combining the id and name of the patient, previous eligibility decision and the justification produced:

```
eligible_list = {
 "Patient" : "f{patient_id}-f{Name}",
 "Decision" : 0/1,
 "Justification" : "<LLM_prompt>"
}
```

## M2-FR1.3 - Justification generation

After the generation of the binary decision by the deterministic motor as the output format preempts a LLM justification must be provided. This process must:

- Feed the whole eligibility list to the LLM
- Feed all the clinical trial logical rules to the LLM
- With this the LLM must output a justification in natural language for the decision, based on these rules.

This process allows for a clear and understandable explanation for the eligibility decision.

## M2-FR1.4 - Persistence of matching

After a patient is considered eligible for a trial, he should be paired with it and this information must be persisted. So it should:

- Store the final eligibility results (decision + justification) in the PostgreSQL table `patient_trial_eligibility` this table connects the patients with the trial.
- Both included and excluded patients must have their decision and justification saved on it. For audit purposes.

## **FR2.1 - Selection of clinical trial**

The system must allow the selection of a clinical trial to be submitted for the calculation of survivability estimations. This process must:

- Provide an interface that presents all clinical trials already submitted
- Allow the selection of one and only one clinical trial to be submitted to these calculations

## **FR2.2 - Retrieve treatment data**

### **FIRST SELECT A DATA RANGE**

After the selection of the a specific clinical trial the system must retrieve relevant patient treatment-related records required for survival analysis. This process must:

- Load, for each patient, the following structured fields from the PostgreSQL database (previously extracted and validated in the "Automated Structured Data Extraction" phase):
  - Diagnosis date
  - Date of first treatment (mapped to First cycle date)
  - Progression date (if present)
  - End date of treatment (if no progression occurred)
  - Death date (if applicable)
- Automatically derive the Status (0/1) required for Kaplan-Meier calculations based on the available data:
  - OS status rules:
    - 1 - if a death date exists
    - 0 - otherwise (censored)
  - PFS status rules:
    - 1 - if a progression date exists
    - 0 - if progression did not occur and only end-of-treatment is available (censored)
- Apply the following deterministic mapping rules between extracted fields on the "Automated Structured Data Extraction" phase and Kaplan-Meier required fields:

Extracted field	Required KM field	Notes
Date of first treatment	First cycle date	Direct mapping
Diagnosis date	Diagnosis date	Direct mapping
Death date	Death date	OS event = 1 if present, else event = 0
End date (from treatments)	End of treatment	Used when no progression; PFS event = 0
-	Progression date	If present → PFS event = 1

### FR2.3 - Compute OS and PFS

After retrieving and mapping all necessary fields in FR2.1, the system must compute the survival metrics Overall Survival (OS) and Progression-Free Survival (PFS) for each patient. This process must:

- Use standardized date formats;
- Apply the following logic for Overall Survival (OS):
  - If death date exists:
    - $OS = (\text{death date} - \text{diagnosis date})$
- Apply the following logic for Progression-Free Survival (PFS):
  - If progression date exists:
    - $PFS = (\text{progression date} - \text{first cycle date})$

### FR2.4 - Apply Kaplan-Meier analysis

After computing OS and PFS for each patient (FR2.2), the system must generate survival estimations using the **Kaplan–Meier method**. This process must:

- Compute the Kaplan–Meier survival function deterministically using the standard formula:

$$S(t) = \prod_{t_i < t} \left(1 - \frac{d_i}{n_i}\right)$$

Where:

- $S(t)$ , probability of surviving beyond time  $t$ .
- $t_i$ , specific time points at which an event occurs.
- $d_i$ , number of events that occur at time  $t_i$ .
- $n_i$ , number of individuals who are still at risk of the event just before time  $t_i$
- Ensure the implementation includes:
  - Grouping of patients by survival time to compute  $d_i$  and  $n_i$ ;

- Proper handling of censored observations (included in the risk set but not counted as events);
- Stepwise survival curves (piecewise constant function).
- The Kaplan-Meier estimates must be outputted into a structured format suitable for visualization (FR2.4).
  - Time points;
  - Survival probabilities;
  - Number at risk;
  - Number of events;
  - Censoring information.

## **FR2.5 - Visualize results**

After computing the Kaplan–Meier survival estimates (FR2.3), the system must generate interactive and exportable visualizations for clinician analysis. This process must:

- Produce visualization-ready survival curves with the following characteristics:
  - Stepwise Kaplan-Meier survival function;
  - Distinct curves for OS and PFS;
  - Markers for censored patients (e.g., "+" symbols);
  - Time displayed on the x-axis and survival probability on the y-axis.
- Allow clinicians to:
  - Select the patient cohort or subgroup to be visualized
  - Toggle between OS and PFS views
  - Overlay multiple curves (e.g., by treatment group or stage)
  - Display underlying survival tables (number at risk per time interval)
- Ensure the graphical output can be:
  - Rendered interactively in the system's dashboard
  - Exported in standard formats (PNG/SVG for images, PDF for reports)
  - Accompanied by summary statistics (median survival, events count, censored count)

## **FR3.1 - Select Diaries for extraction**

After the full load of unstructured clinical diaries into the PostgreSQL database (FR0.1), the system must allow the clinician to select on or multiple diaries to undergo structured data extraction. This process must:

- Retrieve and display all available unstructured diary entries stored in the database.
- Provide an interface allowing the clinician to:
  - Select individual diaries manually;

- Select multiple diaries using batch selection.
- Validate the selection to ensure:
  - Only diaries that have not yet undergone structured extraction (FR3.2-FR3.4) can be selected;
  - Duplicate extractions are prevented;
  - Any missing or corrupted diaries trigger a system warning.
- Forward selected diaries to the structure extraction pipeline (FR3.2-FR3.4) ensuring the system:
  - Records the diary status to "In Extraction" on the database

## FR3.2 - Pre-processing of clinical diary

After the selection of a clinical diary, the system must perform a pre-processing step for the removal of unnecessary information. This process must:

- Remove any unnecessary information to the pipeline, in the case of clinical diaries this is: headers of the file and footers of the file.
- After the end of this preprocessing step a new entry must be persisted on the database referencing the parent entry, making use of the foreign key.

## FR3.3 - Extract structured info

After the selection of relevant unstructured clinical diaries (FR3.1) and its pre-processing, the system must use an LLM-based extraction pipeline to convert free-text diary content into structured patient data. The process must:

- Process each selected diary independently to ensure reproducibility and traceability.
- Each clinical diary must be injected into the LLM through a deterministic zero-shot prompting pipeline, implemented via a Python script.
- For every diary, the script must construct a standardized prompt template, replacing only the free-text diary content to ensure consistent outputs.
- The prompt must include:
  - Clear extraction instructions
  - A fixed schema the model must respect
  - The raw diary text, inserted verbatim in a designated section, avoiding formatting that may leak metadata or context (e.g. <Diary> <\Diary>)
- The full prompt template should be structured like this:

```
You are a clinical information extraction assistant.
Your task is to extract structured clinical data from an unstructured free-
text clinical diary.
```

Follow these rules:

- Extract only the information explicitly present in the diary.
- If a field is missing, set its value to null.
- Do not infer or guess clinical data.
- Output only valid JSON, respecting the specified schema.

Schema (all fields must appear in the output, null if missing):

```
{
 "age_or_birthdate": ...,
 "ecog_ps": ...,
 "diagnosis": ...,
 "diagnosis_date": ...,
 "molecular_status": ...,
 "stage": ...,
 "treatments": {
 "1_treatment": {
 "start_date": ...,
 "end_date": ...
 },
 "2_treatment": {
 "start_date": ...,
 "end_date": ...
 }
 },
 "control": ...
}
```

Clinical diary to process:

```
<DIARY>
{{DIARY_TEXT}}
</DIARY>
```

Return only the JSON object.

- Extract the following clinical parameters from unstructured text:
  - Demographic information:
    - Age and/or Date of birth.
  - Clinical performance metrics:
    - ECOG-PS (eastern Cooperative Oncology Group - Performance Status).
  - Diagnosis-related information

- Primary diagnosis;
- Date of diagnosis;
- Molecular status;
- Cancer stage (TNM or equivalent).
- Treatment information
  - Date of first recorded treatment;
  - Treatments, for each:
    - Start date;
    - End date.
- Control information.
- Ensure robustness of extraction by:
  - Handling missing information gracefully, if a field is not present in the diary, the system must assign the value null;
  - Logging extraction confidence scores or flags for ambiguous fields **LLM AS A JUDGE**
- Produce a structured JSON object for each processed diary in a standardized schema compliant with the database model
- Forward the extracted structured data to the next steps:
  - Normalize text (FR3.4)
  - Validation of extracted parameters (FR3.6)
- Having the parameters extracted and yet to be normalized they must be persisted on a new entry on the database referencing the parent entry, making use of the foreign key.

### FR3.4 - Normalize text

The system must correct typographical and linguistic inconsistencies in the extracted clinical fields. This normalization step is executed **after** the first LLM completes the extraction (FR3.2), and it must be performed individually for each extracted diary to ensure data integrity and traceability. The normalization process must:

- Receive as input the structured JSON generated by the previous extraction stage;
- Apply a second deterministic LLM prompt, using a fixed template, to normalize only the textual fields, while preserving numerical and date fields unchanged
- Ensure consistency in spelling, abbreviations, medical terminology, and capitalization
- Prohibit inference, or correction beyond language normalization.
- Ensure that missing fields remain null, without introducing new data.
- Produce a normalized JSON, following the exact same schema as FR3.2, overwriting only the corrected text fields.
- The full normalization prompt template should be structured like this:

You are a clinical text normalization assistant.  
Your task is to correct typographical and linguistic inconsistencies in the extracted patient data.

Rules:

- Only correct language/format inconsistencies (spelling, abbreviations, capitalization, terminology).
- Do NOT infer missing data or invent information.
- Keep numeric, date, and boolean fields unchanged.
- Output the result strictly in valid JSON, following the schema provided.

Schema (all fields must appear in the output, null if missing):

```
{
 "age_or_birthdate": ...,
 "ecog_ps": ...,
 "diagnosis": ...,
 "diagnosis_date": ...,
 "molecular_status": ...,
 "stage": ...,
 "treatments": {
 "1_treatment": {
 "start_date": ...,
 "end_date": ...
 },
 "2_treatment": {
 "start_date": ...,
 "end_date": ...
 }
 },
 "control": ...
}
```

Here is the extracted JSON from the diary to normalize:

```
<EXTRACTED_JSON>
{{EXTRACTED_JSON}}
</EXTRACTED_JSON>
```

Return only the corrected JSON object.

- With the normalization of the previously extracted fields they again must be persisted in a new entry on the database referencing the parent entry, making use of the foreign key.

### **FR3.5 - Presentation of info**

After the extraction of relevant parameters (FR3.3) and their semantic normalization (FR3.4), the system must present these results to the clinician for review. This process must:

- Display all extracted fields in a interface, organized by:
  - Demographics (Age, date of birth);
  - Performance status (ECOG-PS);
  - Diagnosis information (diagnosis, date of diagnosis, stage, molecular status);
  - Treatments (start date, end date);
  - Control;
- Display as well the raw unstructured text file for an easier correction of these parameters

This process allows for a thorough validation of the parameters extracted.

### **FR3.6 - Validation of extracted parameters**

With the display of these parameters (FR3.5), the system must provide a clinician validation interface to review, edit, and approve the structured data before it is persisted for further analysis. This process must:

- Allow clinicians to:
  - Edit any incorrectly extracted or normalized field;
  - Add missing data if the diary contains information that was not captured
  - Delete erroneously extracted fields or treatments
  - Approve the validated dataset once all required fields are reviewed
- Ensure that validation status is tracked:
  - Each diary receives a validated = TRUE/FALSE flag
  - Record the timestamp for each validation session
  - Store any clinicians edits as a versioned log to ensure auditability
- Only after validation, the structured data should be used for downstream processes:
  - Patient eligibility evaluation for clinical trials (M1-FR1.1-M2-FR1.4);
  - Survival analysis (FR2.1-FR2.5).
- Provide reproducibility and traceability by:
  - Logging the original extracted JSON, normalized JSON, and the final validated JSON
  - Maintaining a history of changes for auditing and quality control.

### **FR3.7 - Store structured profiles**

After extraction (FR3.2), normalization (FR3.3), and clinician validation (FR3.4), the system must persist each structured patient record in the local PostgreSQL database. This process

must:

- Store all validated patient records in a dedicated table `patient_profiles` with a schema that includes:
  - Patient identifier;
  - Demographics (Age/Date of birth);
  - Performance status (ECOG-PS);
  - Diagnosis details (diagnosis, date of diagnosis, stage, molecular status);
  - Treatments (start and end dates)
  - Control
- Along side all the alterations made to the clinical diary: extraction and normalization the validation as well must be persisted on a new entry on the database, again referencing the parent entry, making use of the foreign key.
- In the `patient_profiles` table it is necessary to ensure that:
  - Only clinician-validated data (FR3.4) is stored
  - Fields not present in the diary are stored as NULL
  - All date fields are stored in standardized ISO format
- The unstructured Diary must has its status updated on the database to "Validated" on the database.